



RADEN DAFFA FIRASYAN ADIKUSUMAH

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I am a fresh graduate in Information Systems from Telkom University with a strong interest in Full Stack Development and Artificial Intelligence. Skilled in PHP, Python, RAG, and KG-RAG, with hands-on experience in frontend and backend development. Passionate about building effective digital solutions and continuously developing my technical skills.

Education Level

SMAS BPI 2 - Bandung

Aug 2019 - May 2022

Senior High School, Social Sciences

Telkom University - Bandung

Sep 2022 - Aug 2026

Bachelor of Undergraduate Information System, 3.64/4.00

- Presented the research paper “A Controlled Comparative Study of RAG and KG- RAG for Explainable Asset Maintenance Recommendations” at ICADEIS (International Conference on Advancement in Data Science, E-learning and Information Systems) on June 23–24, 2026, and published it in the IEEE Xplore Digital Library on August 12, 2026.
- Final Project: Developed an LLM-based Decision Support System using RAG and KG- RAG for explainable and traceable asset maintenance recommendations, achieving 4.73/5 overall evaluation, 92.9% procedure match, and 39/39 functional tests passed.

Work Experiences

Arranet (Third Company) - Jakarta

Nov 2024 - Apr 2025

Internship Fullstack Developer

- Developed a nationwide cow farm management web application that integrates real-time IoT systems.
- Designed and built the back-end architecture using Django to handle user data, farm records, and integrate IoT devices.
- Implemented front-end features using HTML, CSS, and JavaScript, ensuring a responsive and user-friendly interface.
- Programmed IoT modules to track and monitor goat activity in real-time through sensor data.
- Integrated IoT data into the web dashboard, enabling users to manage and monitor multiple farm locations across Indonesia.
- Collaborated with the development team to ensure smooth communication between IoT devices and the web platform.
- Tools & Technologies: Python, Django, HTML, CSS, JavaScript, IoT Sensors (e.g., ESP32/Arduino), MQTT, JSON, REST API, Git, Celery, and Redis

Pertamina Hulu Indonesia - Jakarta

Jul 2025 - Aug 2025

Internship AI Fullstack Developer

- Designed and built an AI Chatbot Assistant system using Microservices architecture to centralize information access previously fragmented across over 350 internal company portals.
- Developed the core backend using .NET Core 8 to handle business logic, API management, and data orchestration.
- Designed a relational database schema on Microsoft SQL Server, creating tables for website management, knowledge base documents, and chat session history.
- Built a dedicated microservice using Python and Flask to handle intensive NLP computational tasks and document processing.
- Implemented Retrieval-Augmented Generation (RAG) technology to enhance answer accuracy and minimize hallucination risks by retrieving context from official documents.
- Utilized the Sentence Transformer library to execute text chunking and convert PDF documents into vector embeddings for precise semantic search.
- Developed a standalone Admin Dashboard using Vue.js and TypeScript, enabling administrators to manage integrated websites and

upload PDF knowledge bases.

- Created a portable and reusable Chatbot Widget using Vue.js, designed to be easily embedded into hundreds of existing internal websites without requiring major source code changes.
- Implemented interactive features, such as a feedback loop mechanism (like/dislike buttons), to evaluate and improve the quality of AI responses.
- Delivered a centralized solution that increases employee efficiency in retrieving data and reduces the IT support team's workload regarding repetitive inquiries.
- Executed comprehensive End-to-End integration, ensuring seamless communication between the admin document upload process, AI vector processing, and user chat interface.

Telkom University - BandungJun 2025 - Sep 2025

Contract Fullstack Developer

- Developed a monolithic web application using Laravel (PHP) that integrates a Village Profile portal for public transparency and an E-Commerce module to empower local MSMEs (UMKM).
- Engineered a robust Admin Panel allowing village officials to perform CRUD operations on news articles, multimedia galleries, organizational charts, and dynamic landing page content (welcome messages/running text).
- Built a product catalog system featuring search and category filtering, implementing a "Direct-to-Seller" model by integrating redirections to WhatsApp and Shopee for transaction finalization.
- Implemented an Excel Import feature within the admin dashboard, enabling efficient bulk uploading of product data via standardized templates.
- Designed and developed interactive statistical dashboards to track daily visitor traffic, identify most-viewed products, and monitor content metrics (total news, gallery items).
- Successfully deployed and configured the application on Hostinger, managing domain setup and database optimization for live production use.

Organizational Experience

Central Computer Information - Telkom UniversityDec 2023 - Dec 2024

Media Management Staff

Central Computer Improvement (CCI) is a student organization at Telkom University focused on advancing technology and digital innovation. Established to foster the development of technical skills, CCI brings together students interested in IT and digital solutions through various projects and initiatives. The organization's activities include web development, network management, data research, UI/UX design, and media content management. CCI has been instrumental in creating impactful digital tools, such as student portals and e-voting systems, while providing members with opportunities to enhance their technical expertise and collaborate on innovative projects that contribute to the university's digital transformation.

- Managed content across various platforms, ensuring consistent messaging and alignment with organizational goals.
- Collaborated with cross-functional teams to promote initiatives, events, and digital solutions within the university community.
- Enhanced brand visibility by creating engaging multimedia content and leveraging analytics to optimize media campaigns.
- Gained experience in content planning, digital marketing, and media outreach, contributing to the overall success of the organization's communication efforts.
- Led media strategies for UKM projects, focusing on content creation, social media, and public relations.

Perhimpunan Mahasiswa Bandung - Telkom UniversitySep 2024 - Present

External Division Staff

Perhimpunan Mahasiswa Bandung (PERMIB) is a student organization at Telkom University that brings together students from the Bandung area. The organization aims to strengthen solidarity, foster a sense of community, and promote personal development through various activities and programs. As part of PERMIB, I have engaged in events that enhance communication, leadership, and networking among

students, while also supporting initiatives that represent the interests of students from Bandung. The organization serves as a platform for cultural exchange, collaboration, and creating lasting connections within the campus community.

- Developed and maintained professional relationships with external organizations, including student associations, local businesses, and community partners.
- Coordinated strategic collaborations and external events to enhance the organization's visibility and engagement.
- Managed communication with external stakeholders to strengthen and promote the organization's presence within and outside the university.
- Gained experience in networking, stakeholder management, and project coordination.
- Contributed to outreach initiatives and supported partnership-driven projects, ensuring alignment with organizational goals.

Enterprise Infrastructure Management Laboratorium - Telkom University

Sep 2024 - Present

Human Capital Development Staff

Enterprise Infrastructure Management (EIM) at Telkom University focuses on the comprehensive management and optimization of IT infrastructure at the enterprise level. The program covers various key areas, including network management, cloud computing, security, and the integration of emerging technologies like e-Government data centers and embedded systems. EIM emphasizes sustainable practices through Green IT initiatives and provides students with the opportunity to gain hands-on experience in designing and managing secure, scalable, and efficient infrastructure systems. The program collaborates with industry partners such as Nixtrain and Cisco Academy to enhance students' expertise in network security, data management, and enterprise-level infrastructure solutions.

- Managed social media platforms to improve organizational visibility, creating content that promotes EIM initiatives and engages both internal and external audiences.
- Supported HR activities such as recruitment processes, employee development programs, and performance evaluations to align with organizational goals.
- Contributed to the design and implementation of internal communication campaigns, fostering a collaborative and productive work environment.
- Utilized social media analytics to measure and improve the effectiveness of communication strategies, ensuring alignment with the organization's objectives.

Skills, Achievements, Projects & Other Experience

- **Soft Skills**: Problem Solving, Team Work and Collaboration, Critical Thinking, Creativity, Communication Skills, Time Management, Interrpersonal Skills.
- **Programming Languages**: Python, PHP, C#, TypeScript, HTML, CSS
- **Frameworks & Libraries**: Laravel, Django, .NET Core 8, Flask, Vue.js, Sentence Transformers, React.js
- **AI & Concepts**: RAG, KG-RAG, LLM, NLP, Vector Embeddings, Knowledge Graphs
- **Databases**: MySQL, PostgreSQL, Microsoft SQL Server
- **Development**: GitHub, Swagger UI, Hostinger, Docker, Vercel
- **Design & Modeling**: Figma, Visual Paradigm, Bizagi
- **Infrastructure & Networking**: Oracle Virtual Box, VMware Workstation Pro, Cisco Packet Tracer.
- **Enterprise & Support**: SAP ERP Systems, Manage Engine Service Desk.
- **Language**: Indonesian and English (Intermediate)
- **Workshop** 📍 (2025): 1Ci ERP Low-Code Training for Software Developers
- **Projects** 📍 (2024): Wheelzy Vehicle Rental Website
- **Projects** 📍 (2023): Simple Cashier Application for Stationery.
- **Projects** 📍 (2022): IoT Face Recognition
- **Projects** 📍 (2025): Sistem Pendukung Keputusan Berbasis Web untuk Manajemen Persediaan dan Prediksi Permintaan pada UMKM

